# 山东大学 计算机 学院

# 计算机网络 课程实验报告

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|---|---|---|
| 实验题目：<br>实验五 TCP | | |
| 实验学时：2h | 实验日期： 2023.03.27 | |
| 实验目的：<br>学习掌握 TCP 的相关内容，并查看相应的 TCP 封包。 | | |
| 硬件环境：<br>Windows10 家庭版 | | |
| 软件环境：<br>Wireshark | | |

实验步骤与内容：

实验内容：

实验一：

1. What is the IP address and TCP port number used by the client computer (source) that is transferring the file to gaia.cs.umass.edu? To answer this question, it's probably easiest to select an HTTP message and explore the details of the TCP packet used to carry this HTTP message, using the "details of the selected packet header window" (refer to Figure 2 in the "Getting Started with Wireshark" Lab if you're uncertain about the Wireshark windows.
2. What is the IP address of gaia.cs.umass.edu? On what port number is it sending and receiving TCP segments for this connection?
3. What is the IP address and TCP port number used by your client computer (source) to transfer the file to gaia.cs.umass.edu?

实验二：

5. What is the sequence number of the SYNACK segment sent by gaia.cs.umass.edu to the client computer in reply to the SYN? What is the value of the Acknowledgement field in the SYNACK segment? How did gaia.cs.umass.edu determine that value? What is it in the segment that identifies the segment as a SYNACK segment?
6. What is the sequence number of the TCP segment containing the HTTP POST command? Note that in order to find the POST command, you'll need to dig into the packet content field at the bottom of the Wireshark window, looking for a segment with a "POST" within its DATA field.
7. Consider the TCP segment containing the HTTP POST as the first segment in the TCP connection. What are the sequence numbers of the first six segments in the TCP connection (including the segment containing the HTTP POST)? At what time was each segment sent? When was the ACK for each segment received? Given the difference between when each TCP segment was sent, and when its acknowledgement was received, what is the RTT value for each of the six segments? What is the EstimatedRTT value (see Section 3.5.3, page 242 in text) after the receipt of each ACK? Assume that the value of the EstimatedRTT is equal to the measured RTT for the first segment, and then is computed using the EstimatedRTT equation on page 242 for all subsequent segments.
8. What is the length of each of the first six TCP segments?

9. What is the minimum amount of available buffer space advertised at the received for the entire trace? Does the lack of receiver buffer space ever throttle the sender?
10. Are there any retransmitted segments in the trace file? What did you check for (in the trace) in order to answer this question?
11. How much data does the receiver typically acknowledge in an ACK? Can you identify cases where the receiver is ACKing every other received segment (see Table 3.2 on page 250 in the text).
12. What is the throughput (bytes transferred per unit time) for the TCP connection? Explain how you calculated this value.

实验三：
13. Use the Time-Sequence-Graph(Stevens) plotting tool to view the sequence number versus time plot of segments being sent from the client to the gaia.cs.umass.edu server. Can you identify where TCP's slowstart phase begins and ends, and where congestion avoidance takes over? Comment on ways in which the measured data differs from the idealized behavior of TCP that we've studied in the text.
14. Answer each of two questions above for the trace that you have gathered when you transferred a file from your computer to gaia.cs.umass.edu

实验步骤：
打开 Wireshark，然后根据实验指导书，使用 WireShark 来捕获网络封包，并通过查看每一个包的具体内容来学习相关知识。
实验一：
1. 先通过 nslookup 命令向 DNS 服务器请求 IP 地址，IP 如下图所示：

```
C:\Users\sdu。wzl>nslookup gaia.cs.umass.edu
服务器:  UnKnown
Address:  192.168.254.245

非权威应答:
名称:    gaia.cs.umass.edu
Address:  128.119.245.12
```

之后再查看相应的封包：

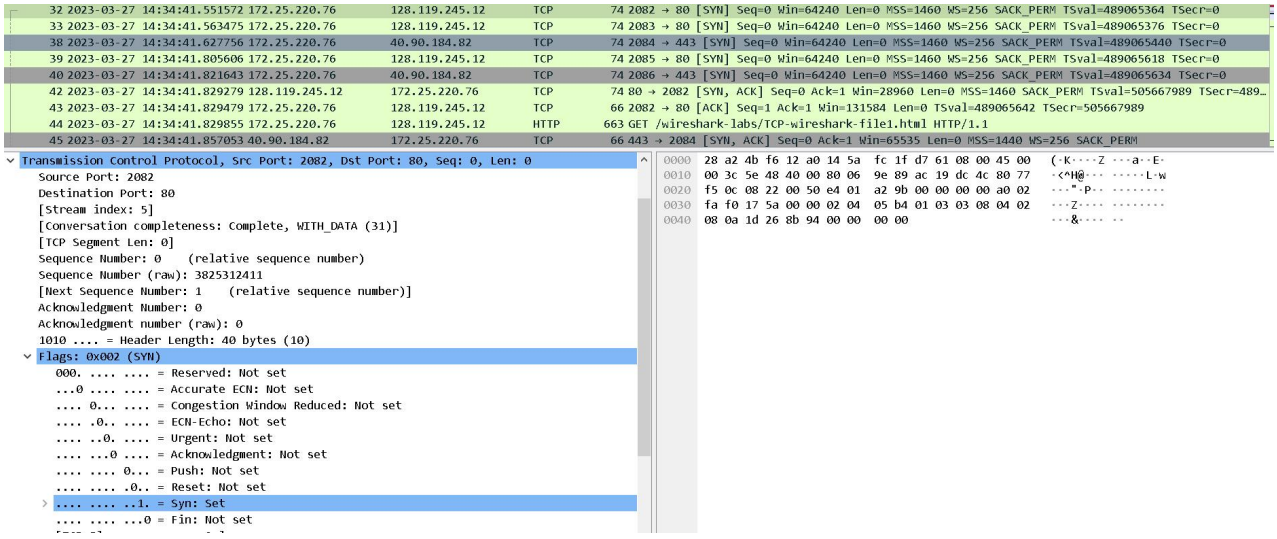
```
299 2023-03-27 14:34:47.770089 172.25.220.76   128.119.245.12   HTTP  1043 POST /wireshark-labs/lab3-1-reply.htm HTTP/1.1  (text/plain
515 2023-03-27 14:34:48.588547 172.25.220.76   128.119.245.12   HTTP   347 POST /wireshark-labs/lab3-1-reply.htm HTTP/1.1  (text/plain
576 2023-03-27 14:34:48.858028 128.119.245.12  172.25.220.76    HTTP   843 HTTP/1.1 200 OK  (text/html)
```

```
Frame 299: 1043 bytes on wire (8344 bits), 1043 bytes captured (8344 bits) on interface \Devi
Ethernet II, Src: LiteonTe_1f:d7:61 (14:5a:fc:1f:d7:61), Dst: JuniperN_f6:12:a0 (28:a2:4b:f6:
Internet Protocol Version 4, Src: 172.25.220.76, Dst: 128.119.245.12
Transmission Control Protocol, Src Port: 2082, Dst Port: 80, Seq: 152689, Ack: 242, Len: 977
[107 Reassembled TCP Segments (153068 bytes): #133(747), #134(1448), #135(1448), #136(1448),
Hypertext Transfer Protocol
MIME Multipart Media Encapsulation, Type: multipart/form-data, Boundary: "----WebKitFormBound
```

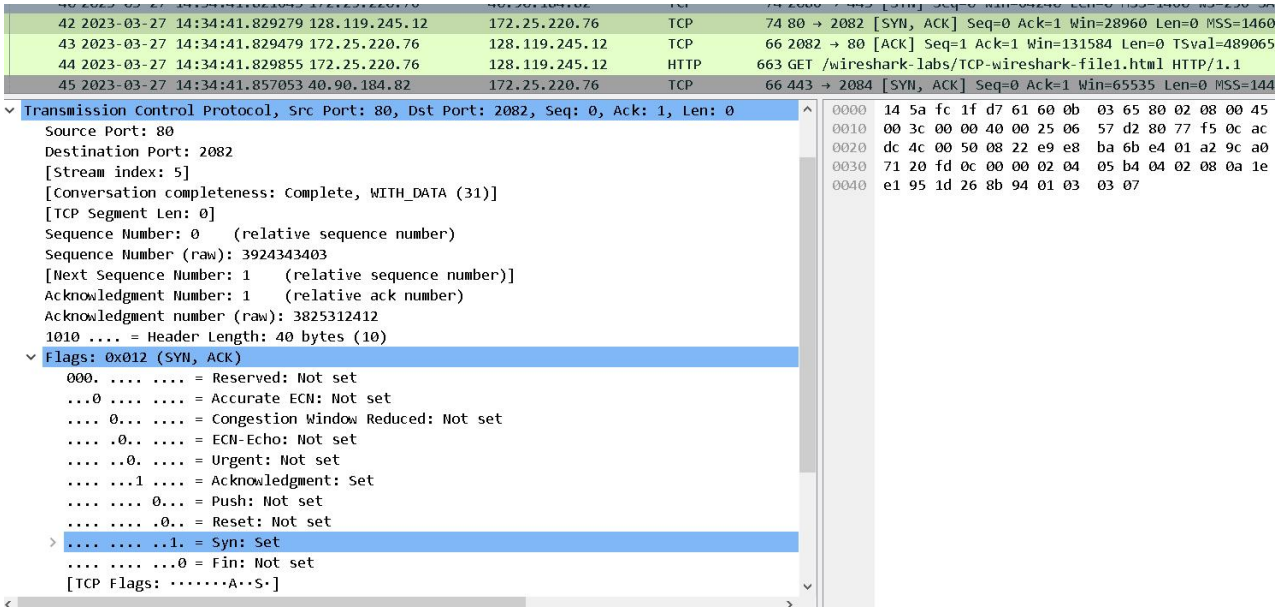
```
0000  28 a2 4b f6 12 a0 14 5a  fc 1f d7 61 08 00 45 00
0010  04 05 5e ba 40 00 80 06  9a 4e ac 19 dc 4c 80 77
0020  f5 0c 08 22 00 50 e4 03  f7 0c e9 e8 bb 5d 80 19
0030  02 01 2e 56 00 00 01 01  08 0a 1d 26 a3 de 1e 23
0040  f8 b1 65 20 51 75 65 65  6e 27 73 20 73 68 72 69
0050  6c 6c 20 63 72 69 65 73  20 74 6f 20 74 68 65 20
0060  76 6f 69 63 65 20 6f 66  20 74 68 65 20 73 68 65
0070  70 68 65 72 64 0d 0a 62  6f 79 2d 2d 61 6e 64 20
0080  74 68 65 20 73 6e 65 65  7a 65 20 6f 66 20 74 68
0090  65 20 62 61 62 79 2c 20  74 68 65 20 73 68 72 69
00a0  65 6b 20 6f 66 20 74 68  65 20 47 72 79 70 68 6f
00b0  6e 2c 20 61 6e 64 0d 0a  61 6c 6c 20 74 68 65 20
00c0  6f 74 68 65 72 20 71 75  65 65 72 20 6e 6f 69 73
00d0  65 73 2c 20 77 6f 75 6c  64 20 63 68 61 6e 67 65
00e0  20 28 73 68 65 20 6b 6e  65 77 29 20 74 6f 20 74
00f0  68 65 0d 0a 63 6f 6e 66  75 73 65 64 20 63 6c 61
0100  6d 6f 75 72 20 6f 66 20  74 68 65 20 62 75 73 79
0110  20 66 61 72 6d 2d 79 61  72 64 2d 2d 77 68 69 6c
0120  65 20 74 68 65 20 6c 6f  77 69 6e 67 20 6f 66 20
0130  74 68 65 0d 0a 63 61 74  74 6c 65 20 69 6e 20 74
```

源 IP 地址是 172.25.220.76 ，源端口是 2082，目的端口号是 80。
2. gaia.cs.umass.edu 的 IP 地址为 128.119.245.12，连接的端口号是 80。
3. 客户端的 IP 地址为 172.25.220.76，TCP 的端口号是 80。

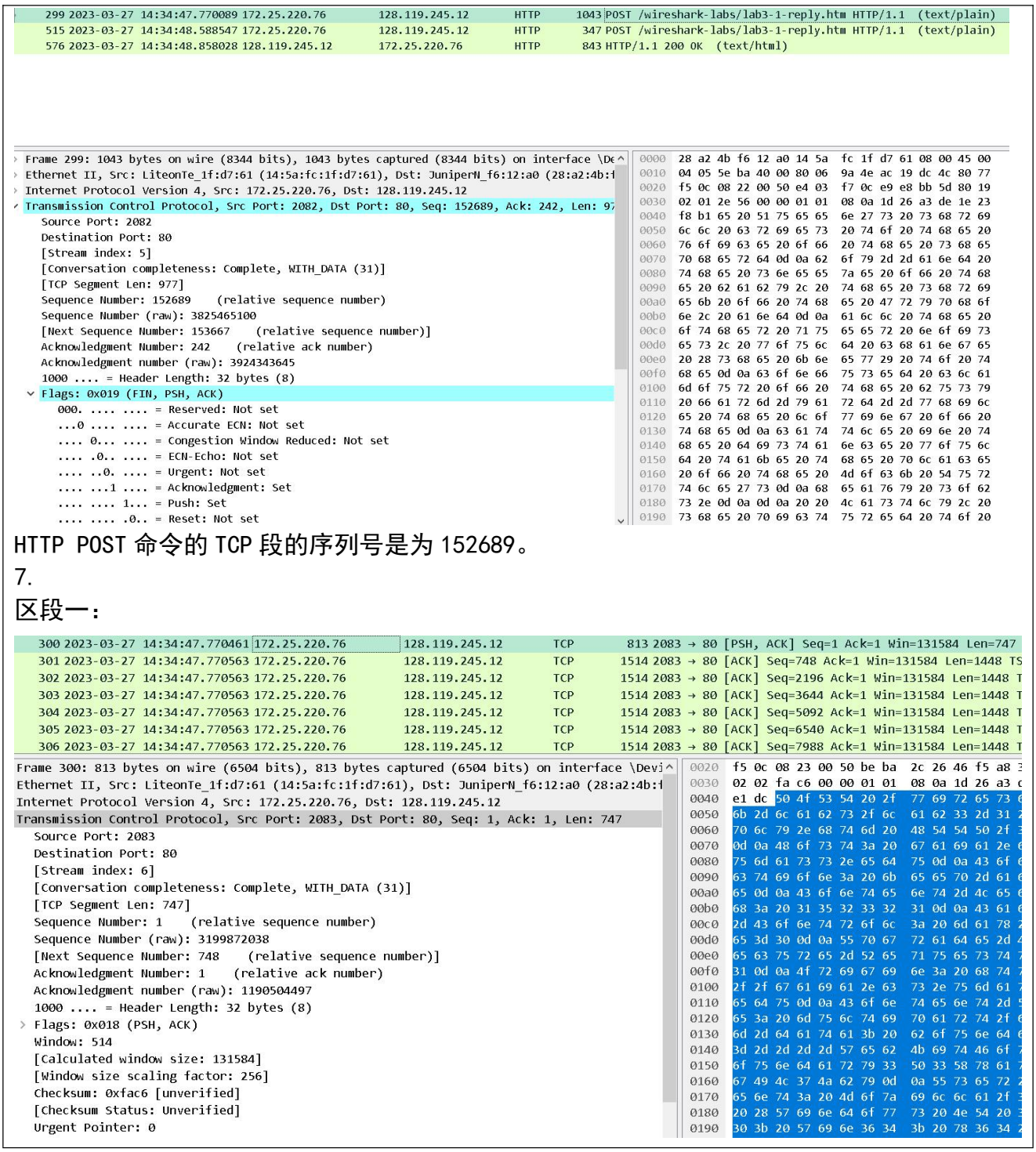
实验二：
4.




通过上图可见 Seq 为 0，通过设置 Flags 为 0x002，来使得 Syn 为 1。
5.



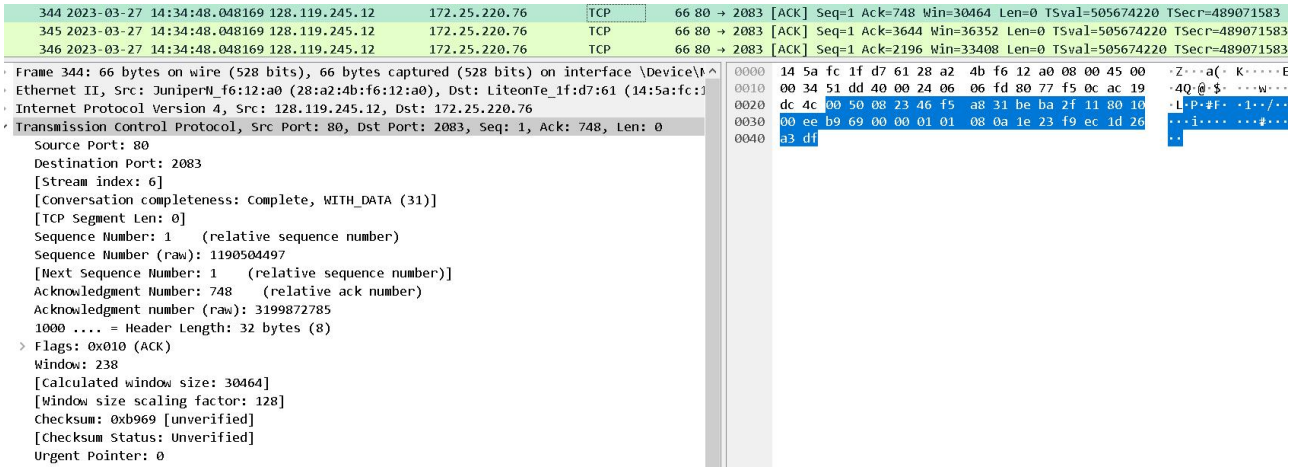

通过上图可见 Seq 为 0，同时 ACK 为 1，通过设置 Flags 为 0x012 来使得 ACK 为 1，Syn 为 1。
6.



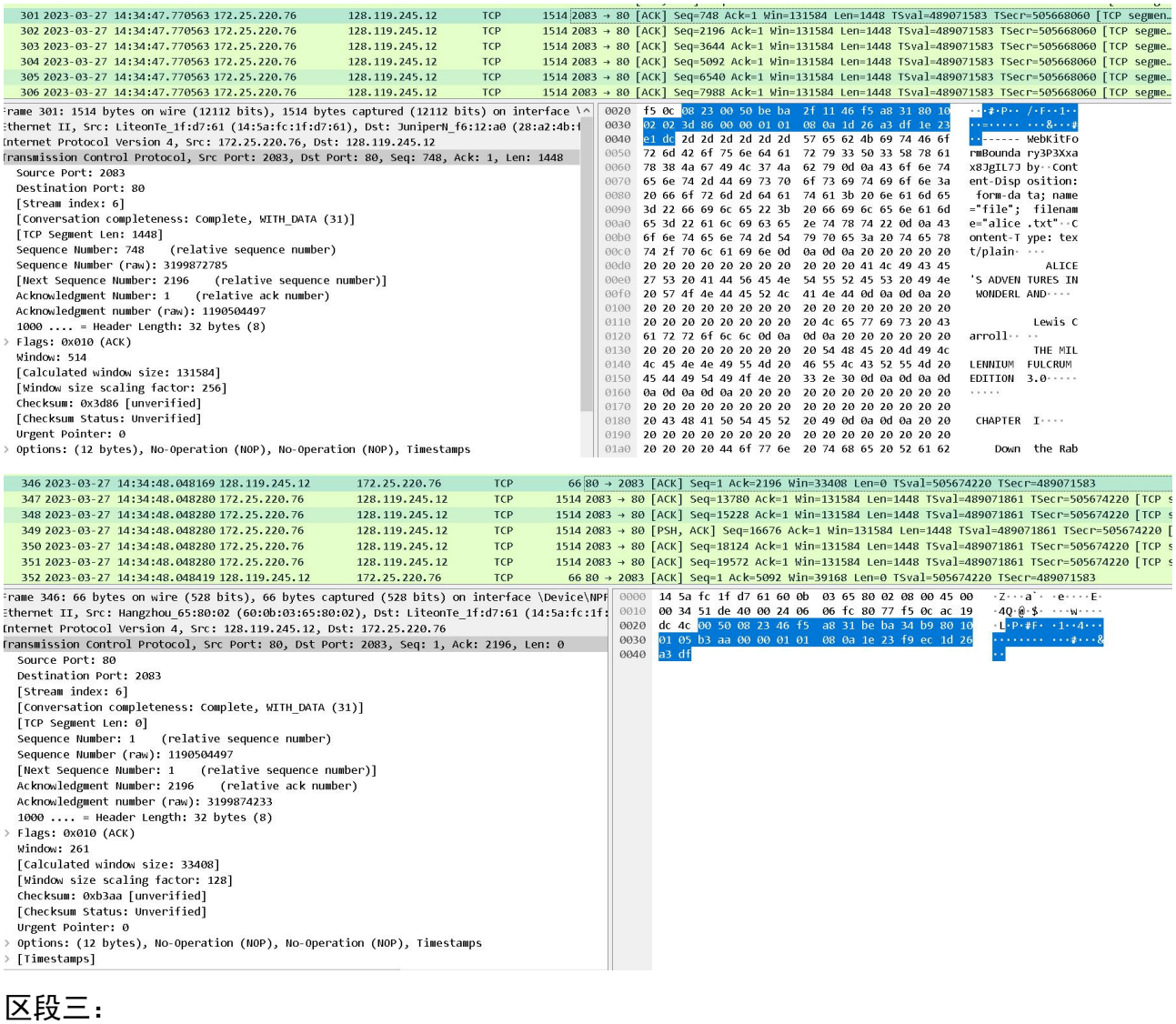

HTTP POST 命令的 TCP 段的序列号是为 152689。

7.

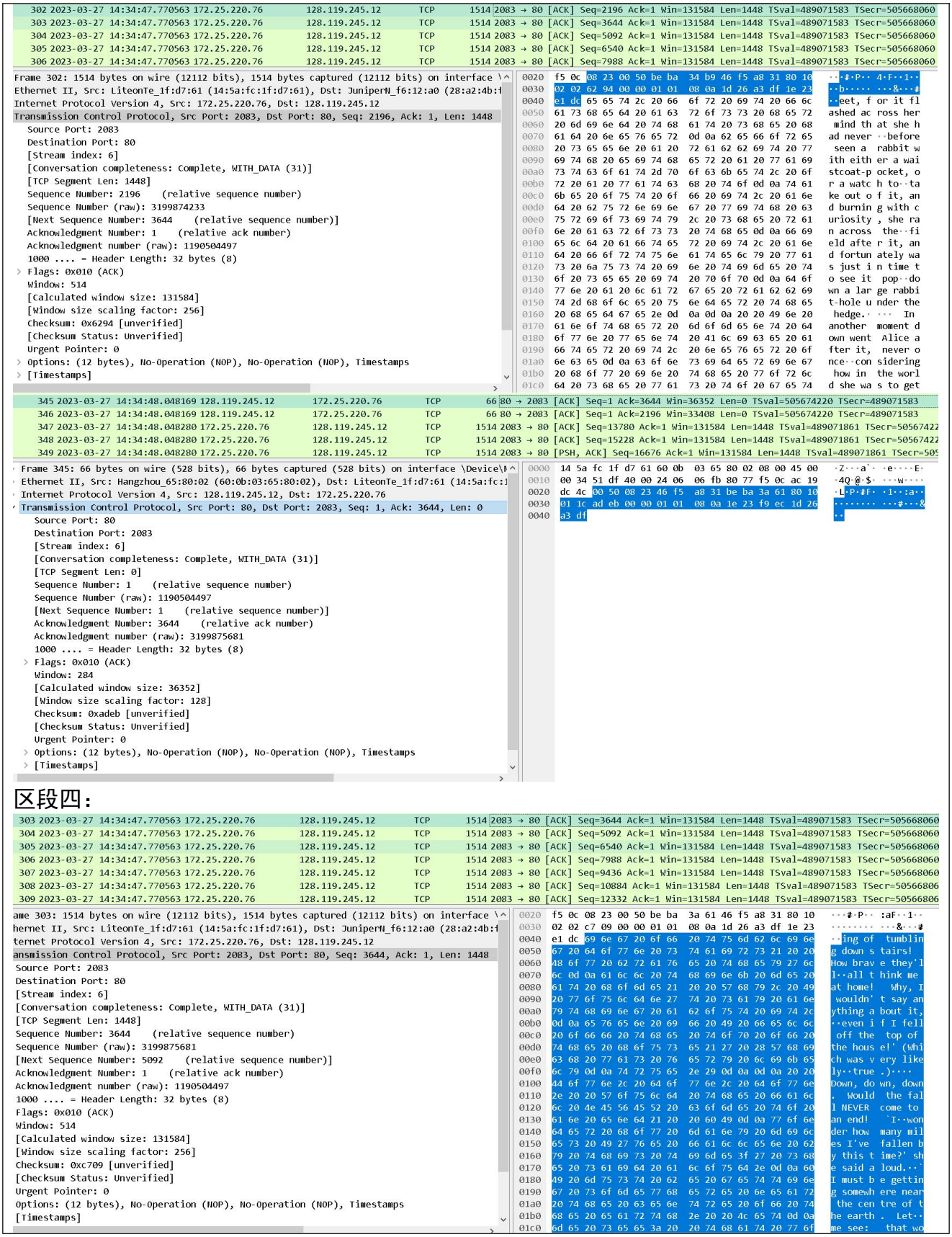
区段一：


344 2023-03-27 14:34:48.048169 128.119.245.12 172.25.220.76 TCP 66 80 → 2083 [ACK] Seq=1 Ack=748 Win=30464 Len=0 TSval=505674220 TSecr=489071583
345 2023-03-27 14:34:48.048169 128.119.245.12 172.25.220.76 TCP 66 80 → 2083 [ACK] Seq=1 Ack=3644 Win=36352 Len=0 TSval=505674220 TSecr=489071583
346 2023-03-27 14:34:48.048169 128.119.245.12 172.25.220.76 TCP 66 80 → 2083 [ACK] Seq=1 Ack=2196 Win=33408 Len=0 TSval=505674220 TSecr=489071583
Frame 344: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\
Ethernet II, Src: JuniperN_f6:12:a0 (28:a2:4b:f6:12:a0), Dst: LiteonTe_1f:d7:61 (14:5a:fc:
Internet Protocol Version 4, Src: 128.119.245.12, Dst: 172.25.220.76
Transmission Control Protocol, Src Port: 80, Dst Port: 2083, Seq: 1, Ack: 748, Len: 0
Source Port: 80
Destination Port: 2083
[Stream index: 6]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 0]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 1190504497
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 748 (relative ack number)
Acknowledgment number (raw): 3199872785
1000 .... = Header Length: 32 bytes (8)
Flags: 0x010 (ACK)
Window: 238
[Calculated window size: 30464]
[Window size scaling factor: 128]
Checksum: 0xb969 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
0000 14 5a fc 1f d7 61 28 a2 4b f6 12 a0 08 00 45 00
0010 00 34 51 dd 40 00 24 06 fd 80 77 f5 0c ac 19
0020 dc 4c 00 50 08 23 46 f5 a8 31 be ba 2f 11 80 10
0030 00 ee b9 69 00 00 01 01 08 0a 1e 23 f9 ec 1d 26
0040 a3 df

## 区段二：


301 2023-03-27 14:34:47.770563 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=748 Ack=1 Win=131584 Len=1448 TSval=489071583 TSecr=505668060 [TCP segmen
302 2023-03-27 14:34:47.770563 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=2196 Ack=1 Win=131584 Len=1448 TSval=489071583 TSecr=505668060 [TCP segme
303 2023-03-27 14:34:47.770563 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=3644 Ack=1 Win=131584 Len=1448 TSval=489071583 TSecr=505668060 [TCP segme
304 2023-03-27 14:34:47.770563 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=5092 Ack=1 Win=131584 Len=1448 TSval=489071583 TSecr=505668060 [TCP segme
305 2023-03-27 14:34:47.770563 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=6540 Ack=1 Win=131584 Len=1448 TSval=489071583 TSecr=505668060 [TCP segme
306 2023-03-27 14:34:47.770563 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=7988 Ack=1 Win=131584 Len=1448 TSval=489071583 TSecr=505668060 [TCP segme
rame 301: 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits) on interface \
thernet II, Src: LiteonTe_1f:d7:61 (14:5a:fc:1f:d7:61), Dst: JuniperN_f6:12:a0 (28:a2:4b:
nternet Protocol Version 4, Src: 172.25.220.76, Dst: 128.119.245.12
ransmission Control Protocol, Src Port: 2083, Dst Port: 80, Seq: 748, Ack: 1, Len: 1448
Source Port: 2083
Destination Port: 80
[Stream index: 6]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 1448]
Sequence Number: 748 (relative sequence number)
Sequence Number (raw): 3199872785
[Next Sequence Number: 2196 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 1190504497
1000 .... = Header Length: 32 bytes (8)
Flags: 0x010 (ACK)
Window: 514
[Calculated window size: 131584]
[Window size scaling factor: 256]
Checksum: 0x3d86 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
0020 f5 0c 08 23 00 50 be ba 2f 11 46 f5 a8 31 80 10
0030 02 02 3d 86 00 00 01 01 08 0a 1d 26 a3 df 1e 23
0040 e1 dc 2d 2d 2d 2d 2d 2d 57 65 62 4b 69 74 46 6f
0050 72 6d 42 6f 75 6e 64 61 72 79 33 50 33 58 78 61
0060 78 38 4a 67 49 4c 37 4a 62 79 0d 0a 43 6f 6e 74
0070 65 6e 74 2d 44 69 73 70 6f 73 69 74 69 6f 6e 3a
0080 20 66 6f 72 6d 2d 64 61 74 61 3b 20 6e 61 6d 65
0090 3d 22 66 69 6c 65 22 3b 20 66 69 6c 65 6e 61 6d
00a0 65 3d 22 61 6c 69 63 65 2e 74 78 74 22 0d 0a 43
00b0 6f 6e 74 65 6e 74 2d 54 79 70 65 3a 20 74 65 78
00c0 74 2f 70 6c 61 69 6e 0d 0a 0d 0a 20 20 20 20 20
00d0 20 20 20 20 20 20 20 20 20 20 20 20 41 4c 49 43 45
00e0 27 53 20 41 44 56 45 4e 54 55 52 45 53 20 49 4e
00f0 20 57 4f 4e 44 45 52 4c 41 4e 44 0d 0a 0d 0a 20
0100 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20
0110 20 20 20 20 20 20 20 20 20 4c 65 77 69 73 20 43
0120 61 72 72 6f 6c 6c 0d 0a 0d 0a 20 20 20 20 20 20
0130 20 20 20 20 20 20 20 20 20 54 48 45 20 4d 49 4c
0140 4c 45 4e 4e 49 55 4d 20 46 55 4c 43 52 55 4d 20
0150 45 44 49 54 49 4f 4e 20 33 2e 30 0d 0a 0d 0a 0d
0160 0a 0d 0a 0d 0a 20 20 20 20 20 20 20 20 20 20 20
0170 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20
0180 20 43 48 41 50 54 45 52 20 49 0d 0a 0d 0a 20 20
0190 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20
01a0 20 20 20 20 44 6f 77 6e 20 74 68 65 20 52 61 62
346 2023-03-27 14:34:48.048169 128.119.245.12 172.25.220.76 TCP 66 80 → 2083 [ACK] Seq=1 Ack=2196 Win=33408 Len=0 TSval=505674220 TSecr=489071583
347 2023-03-27 14:34:48.048280 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=13780 Ack=1 Win=131584 Len=1448 TSval=489071861 TSecr=505674220 [TCP s
348 2023-03-27 14:34:48.048280 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=15228 Ack=1 Win=131584 Len=1448 TSval=489071861 TSecr=505674220 [TCP s
349 2023-03-27 14:34:48.048280 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [PSH, ACK] Seq=16676 Ack=1 Win=131584 Len=1448 TSval=489071861 TSecr=505674220 [
350 2023-03-27 14:34:48.048280 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=18124 Ack=1 Win=131584 Len=1448 TSval=489071861 TSecr=505674220 [TCP s
351 2023-03-27 14:34:48.048280 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=19572 Ack=1 Win=131584 Len=1448 TSval=489071861 TSecr=505674220 [TCP s
352 2023-03-27 14:34:48.048419 128.119.245.12 172.25.220.76 TCP 66 80 → 2083 [ACK] Seq=1 Ack=5092 Win=39168 Len=0 TSval=505674220 TSecr=489071583
rame 346: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF
thernet II, Src: Hangzhou_65:80:02 (60:0b:03:65:80:02), Dst: LiteonTe_1f:d7:61 (14:5a:fc:1f:
nternet Protocol Version 4, Src: 128.119.245.12, Dst: 172.25.220.76
ransmission Control Protocol, Src Port: 80, Dst Port: 2083, Seq: 1, Ack: 2196, Len: 0
Source Port: 80
Destination Port: 2083
[Stream index: 6]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 0]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 1190504497
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 2196 (relative ack number)
Acknowledgment number (raw): 3199874233
1000 .... = Header Length: 32 bytes (8)
Flags: 0x010 (ACK)
Window: 261
[Calculated window size: 33408]
[Window size scaling factor: 128]
Checksum: 0xb3aa [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
[Timestamps]
0000 14 5a fc 1f d7 61 60 0b 03 65 80 02 08 00 45 00
0010 00 34 51 de 40 00 24 06 06 fc 80 77 f5 0c ac 19
0020 dc 4c 00 50 08 23 46 f5 a8 31 be ba 34 b9 80 10
0030 01 05 b3 aa 00 00 01 01 08 0a 1e 23 f9 ec 1d 26
0040 a3 df

## 区段三：


302 2023-03-27 14:34:47.770563 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=2196 Ack=1 Win=131584 Len=1448 TSval=489071583 TSecr=505668060
303 2023-03-27 14:34:47.770563 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=3644 Ack=1 Win=131584 Len=1448 TSval=489071583 TSecr=505668060
304 2023-03-27 14:34:47.770563 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=5092 Ack=1 Win=131584 Len=1448 TSval=489071583 TSecr=505668060
305 2023-03-27 14:34:47.770563 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=6540 Ack=1 Win=131584 Len=1448 TSval=489071583 TSecr=505668060
306 2023-03-27 14:34:47.770563 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=7988 Ack=1 Win=131584 Len=1448 TSval=489071583 TSecr=505668060
Frame 302: 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits) on interface \
Ethernet II, Src: LiteonTe_1f:d7:61 (14:5a:fc:1f:d7:61), Dst: JuniperN_f6:12:a0 (28:a2:4b:f
Internet Protocol Version 4, Src: 172.25.220.76, Dst: 128.119.245.12
Transmission Control Protocol, Src Port: 2083, Dst Port: 80, Seq: 2196, Ack: 1, Len: 1448
Source Port: 2083
Destination Port: 80
[Stream index: 6]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 1448]
Sequence Number: 2196 (relative sequence number)
Sequence Number (raw): 3199874233
[Next Sequence Number: 3644 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 1190504497
1000 .... = Header Length: 32 bytes (8)
Flags: 0x010 (ACK)
Window: 514
[Calculated window size: 131584]
[Window size scaling factor: 256]
Checksum: 0x6294 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
[Timestamps]
0020 f5 0c 08 23 00 50 be ba 34 b9 46 f5 a8 31 80 10 ···#·P·· 4·F··1··
0030 02 02 62 94 00 00 01 01 08 0a 1d 26 a3 df 1e 23 ··b····· ···&···#
0040 e1 dc 65 65 74 2c 20 66 6f 72 20 69 74 20 66 6c ··eet, f or it fl
0050 61 73 68 65 64 20 61 63 72 6f 73 73 20 68 65 72 ashed ac ross her
0060 20 6d 69 6e 64 20 74 68 61 74 20 73 68 65 20 68 mind th at she h
0070 61 64 20 6e 65 76 65 72 0d 0a 62 65 66 6f 72 65 ad never ··before
0080 20 73 65 65 6e 20 61 20 72 61 62 62 69 74 20 77 seen a rabbit w
0090 69 74 68 20 65 69 74 68 65 72 20 61 20 77 61 69 ith eith er a wai
00a0 73 74 63 6f 61 74 2d 70 6f 63 6b 65 74 2c 20 6f stcoat-p ocket, o
00b0 72 20 61 20 77 61 74 63 68 20 74 6f 0d 0a 74 61 r a watc h to··ta
00c0 6b 65 20 6f 75 74 20 6f 66 20 69 74 2c 20 61 6e ke out o f it, an
00d0 64 20 62 75 72 6e 69 6e 67 20 77 69 74 68 20 63 d burnin g with c
00e0 75 72 69 6f 73 69 74 79 2c 20 73 68 65 20 72 61 uriosity , she ra
00f0 6e 20 61 63 72 6f 73 73 20 74 68 65 0d 0a 66 69 n across the··fi
0100 65 6c 64 20 61 66 74 65 72 20 69 74 2c 20 61 6e eld afte r it, an
0110 64 20 66 6f 72 74 75 6e 61 74 65 6c 79 20 77 61 d fortun ately wa
0120 73 20 6a 75 73 74 20 69 6e 20 74 69 6d 65 20 74 s just i n time t
0130 6f 20 73 65 65 20 69 74 20 70 6f 70 0d 0a 64 6f o see it pop··do
0140 77 6e 20 61 20 6c 61 72 67 65 20 72 61 62 62 69 wn a lar ge rabbi
0150 74 2d 68 6f 6c 65 20 75 6e 64 65 72 20 74 68 65 t-hole u nder the
0160 20 68 65 64 67 65 2e 0d 0a 0d 0a 20 20 49 6e 20 hedge.· ··· In
0170 61 6e 6f 74 68 65 72 20 6d 6f 6d 65 6e 74 20 64 another moment d
0180 6f 77 6e 20 77 65 6e 74 20 41 6c 69 63 65 20 61 own went Alice a
0190 66 74 65 72 20 69 74 2c 20 6e 65 76 65 72 20 6f fter it, never o
01a0 6e 63 65 0d 0a 63 6f 6e 73 69 64 65 72 69 6e 67 nce··con sidering
01b0 20 68 6f 77 20 69 6e 20 74 68 65 20 77 6f 72 6c how in the worl
01c0 64 20 73 68 65 20 77 61 73 20 74 6f 20 67 65 74 d she wa s to get
345 2023-03-27 14:34:48.048169 128.119.245.12 172.25.220.76 TCP 66 80 → 2083 [ACK] Seq=1 Ack=3644 Win=36352 Len=0 TSval=505674220 TSecr=489071583
346 2023-03-27 14:34:48.048169 128.119.245.12 172.25.220.76 TCP 66 80 → 2083 [ACK] Seq=1 Ack=2196 Win=33408 Len=0 TSval=505674220 TSecr=489071583
347 2023-03-27 14:34:48.048280 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=13780 Ack=1 Win=131584 Len=1448 TSval=489071861 TSecr=50567422
348 2023-03-27 14:34:48.048280 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=15228 Ack=1 Win=131584 Len=1448 TSval=489071861 TSecr=50567422
349 2023-03-27 14:34:48.048280 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [PSH, ACK] Seq=16676 Ack=1 Win=131584 Len=1448 TSval=489071861 TSecr=505
Frame 345: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\N
Ethernet II, Src: Hangzhou_65:80:02 (60:0b:03:65:80:02), Dst: LiteonTe_1f:d7:61 (14:5a:fc:1
Internet Protocol Version 4, Src: 128.119.245.12, Dst: 172.25.220.76
Transmission Control Protocol, Src Port: 80, Dst Port: 2083, Seq: 1, Ack: 3644, Len: 0
Source Port: 80
Destination Port: 2083
[Stream index: 6]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 0]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 1190504497
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 3644 (relative ack number)
Acknowledgment number (raw): 3199875681
1000 .... = Header Length: 32 bytes (8)
Flags: 0x010 (ACK)
Window: 284
[Calculated window size: 36352]
[Window size scaling factor: 128]
Checksum: 0xadeb [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
[Timestamps]
0000 14 5a fc 1f d7 61 60 0b 03 65 80 02 08 00 45 00 ·Z···a`· ·e····E·
0010 00 34 51 df 40 00 24 06 06 fb 80 77 f5 0c ac 19 ·4Q·@·$· ···w····
0020 dc 4c 00 50 08 23 46 f5 a8 31 be ba 3a 61 80 10 ·L·P·#F· ·1··:a··
0030 01 1c ad eb 00 00 01 01 08 0a 1e 23 f9 ec 1d 26 ········ ···#···&
0040 a3 df ··


## 区段四：


303 2023-03-27 14:34:47.770563 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=3644 Ack=1 Win=131584 Len=1448 TSval=489071583 TSecr=505668060
304 2023-03-27 14:34:47.770563 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=5092 Ack=1 Win=131584 Len=1448 TSval=489071583 TSecr=505668060
305 2023-03-27 14:34:47.770563 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=6540 Ack=1 Win=131584 Len=1448 TSval=489071583 TSecr=505668060
306 2023-03-27 14:34:47.770563 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=7988 Ack=1 Win=131584 Len=1448 TSval=489071583 TSecr=505668060
307 2023-03-27 14:34:47.770563 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=9436 Ack=1 Win=131584 Len=1448 TSval=489071583 TSecr=505668060
308 2023-03-27 14:34:47.770563 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=10884 Ack=1 Win=131584 Len=1448 TSval=489071583 TSecr=505668060
309 2023-03-27 14:34:47.770563 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=12332 Ack=1 Win=131584 Len=1448 TSval=489071583 TSecr=505668060
ame 303: 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits) on interface \
hernet II, Src: LiteonTe_1f:d7:61 (14:5a:fc:1f:d7:61), Dst: JuniperN_f6:12:a0 (28:a2:4b:f
ternet Protocol Version 4, Src: 172.25.220.76, Dst: 128.119.245.12
ansmission Control Protocol, Src Port: 2083, Dst Port: 80, Seq: 3644, Ack: 1, Len: 1448
Source Port: 2083
Destination Port: 80
[Stream index: 6]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 1448]
Sequence Number: 3644 (relative sequence number)
Sequence Number (raw): 3199875681
[Next Sequence Number: 5092 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 1190504497
1000 .... = Header Length: 32 bytes (8)
Flags: 0x010 (ACK)
Window: 514
[Calculated window size: 131584]
[Window size scaling factor: 256]
Checksum: 0xc709 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
[Timestamps]
0020 f5 0c 08 23 00 50 be ba 3a 61 46 f5 a8 31 80 10 ···#·P·· :aF··1··
0030 02 02 c7 09 00 00 01 01 08 0a 1d 26 a3 df 1e 23 ········ ···&···#
0040 e1 dc 69 6e 67 20 6f 66 20 74 75 6d 62 6c 69 6e ··ing of tumblin
0050 67 20 64 6f 77 6e 20 73 74 61 69 72 73 21 20 20 g down s tairs!
0060 48 6f 77 20 62 72 61 76 65 20 74 68 65 79 27 6c How brav e they'l
0070 6c 0d 0a 61 6c 6c 20 74 68 69 6e 6b 20 6d 65 20 l··all t hink me
0080 61 74 20 68 6f 6d 65 21 20 20 57 68 79 2c 20 49 at home! Why, I
0090 20 77 6f 75 6c 64 6e 27 74 20 73 61 79 20 61 6e wouldn' t say an
00a0 79 74 68 69 6e 67 20 61 62 6f 75 74 20 69 74 2c ything a bout it,
00b0 0d 0a 65 76 65 6e 20 69 66 20 49 20 66 65 6c 6c ··even i f I fell
00c0 20 6f 66 66 20 74 68 65 20 74 6f 70 20 6f 66 20 off the top of
00d0 74 68 65 20 68 6f 75 73 65 21 27 20 28 57 68 69 the hous e!' (Whi
00e0 63 68 20 77 61 73 20 76 65 72 79 20 6c 69 6b 65 ch was v ery like
00f0 6c 79 0d 0a 74 72 75 65 2e 29 0d 0a 0d 0a 20 20 ly··true .)····
0100 44 6f 77 6e 2c 20 64 6f 77 6e 2c 20 64 6f 77 6e Down, do wn, down
0110 2e 20 20 57 6f 75 6c 64 20 74 68 65 20 66 61 6c . Would the fal
0120 6c 20 4e 45 56 45 52 20 63 6f 6d 65 20 74 6f 20 l NEVER come to
0130 61 6e 20 65 6e 64 21 20 20 60 49 0d 0a 77 6f 6e an end! `I··won
0140 64 65 72 20 68 6f 77 20 6d 61 6e 79 20 6d 69 6c der how many mil
0150 65 73 20 49 27 76 65 20 66 61 6c 6c 65 6e 20 62 es I've fallen b
0160 79 20 74 68 69 73 20 74 69 6d 65 3f 27 20 73 68 y this t ime?' sh
0170 65 20 73 61 69 64 20 61 6c 6f 75 64 2e 0d 0a 60 e said a loud.··`
0180 49 20 6d 75 73 74 20 62 65 20 67 65 74 74 69 6e I must b e gettin
0190 67 20 73 6f 6d 65 77 68 65 72 65 20 6e 65 61 72 g somewh ere near
01a0 20 74 68 65 20 63 65 6e 74 72 65 20 6f 66 20 74 the cen tre of t
01b0 68 65 20 65 61 72 74 68 2e 20 20 4c 65 74 0d 0a he earth . Let··
01c0 6d 65 20 73 65 65 3a 20 20 74 68 61 74 20 77 6f me see: that wo

```
352 2023-03-27 14:34:48.048419 128.119.245.12  172.25.220.76   TCP    66 80 → 2083 [ACK] Seq=1 Ack=5092 Win=39168 Len=0
353 2023-03-27 14:34:48.048440 172.25.220.76   128.119.245.12  TCP  1514 2083 → 80 [ACK] Seq=21020 Ack=1 Win=131584 Len=
354 2023-03-27 14:34:48.048440 172.25.220.76   128.119.245.12  TCP  1514 2083 → 80 [ACK] Seq=22468 Ack=1 Win=131584 Len=
355 2023-03-27 14:34:48.048490 128.119.245.12  172.25.220.76   TCP    66 80 → 2083 [ACK] Seq=1 Ack=7988 Win=45056 Len=0
```

```
ame 345: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\N
thernet II, Src: Hangzhou_65:80:02 (60:0b:03:65:80:02), Dst: LiteonTe_1f:d7:61 (14:5a:fc:1
nternet Protocol Version 4, Src: 128.119.245.12, Dst: 172.25.220.76
ansmission Control Protocol, Src Port: 80, Dst Port: 2083, Seq: 1, Ack: 3644, Len: 0
 Source Port: 80
 Destination Port: 2083
 [Stream index: 6]
 [Conversation completeness: Complete, WITH_DATA (31)]
 [TCP Segment Len: 0]
 Sequence Number: 1    (relative sequence number)
 Sequence Number (raw): 1190504497
 [Next Sequence Number: 1    (relative sequence number)]
 Acknowledgment Number: 3644    (relative ack number)
 Acknowledgment number (raw): 3199875681
 1000 .... = Header Length: 32 bytes (8)
 Flags: 0x010 (ACK)
 Window: 284
 [Calculated window size: 36352]
 [Window size scaling factor: 128]
 Checksum: 0xadeb [unverified]
 [Checksum Status: Unverified]
 Urgent Pointer: 0
 Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
 [Timestamps]
```

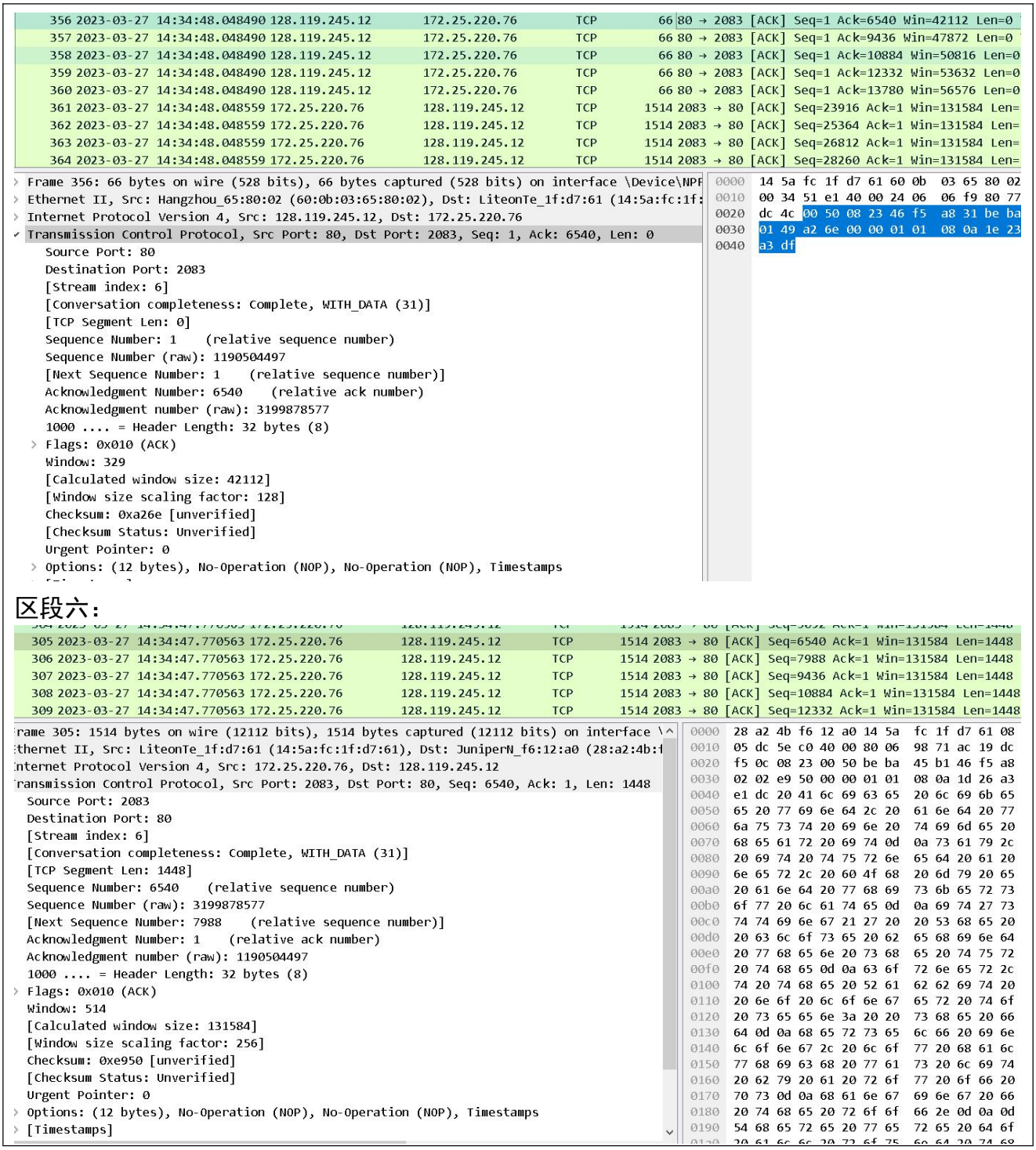
```
0000  14 5a fc 1f d7 61 60 0b  03 65 80 02
0010  00 34 51 df 40 00 24 06  06 fb 80 77
0020  dc 4c 00 50 08 23 46 f5  a8 31 be ba
0030  01 1c ad eb 00 00 01 01  08 0a 1e 23
0040  a3 df
```

## 区段五：

```
304 2023-03-27 14:34:47.770563 172.25.220.76   128.119.245.12  TCP  1514 2083 → 80 [ACK] Seq=5092 Ack=1 Win=131584 Len=1448
305 2023-03-27 14:34:47.770563 172.25.220.76   128.119.245.12  TCP  1514 2083 → 80 [ACK] Seq=6540 Ack=1 Win=131584 Len=1448
306 2023-03-27 14:34:47.770563 172.25.220.76   128.119.245.12  TCP  1514 2083 → 80 [ACK] Seq=7988 Ack=1 Win=131584 Len=1448
307 2023-03-27 14:34:47.770563 172.25.220.76   128.119.245.12  TCP  1514 2083 → 80 [ACK] Seq=9436 Ack=1 Win=131584 Len=1448
308 2023-03-27 14:34:47.770563 172.25.220.76   128.119.245.12  TCP  1514 2083 → 80 [ACK] Seq=10884 Ack=1 Win=131584 Len=1448
309 2023-03-27 14:34:47.770563 172.25.220.76   128.119.245.12  TCP  1514 2083 → 80 [ACK] Seq=12332 Ack=1 Win=131584 Len=1448
```

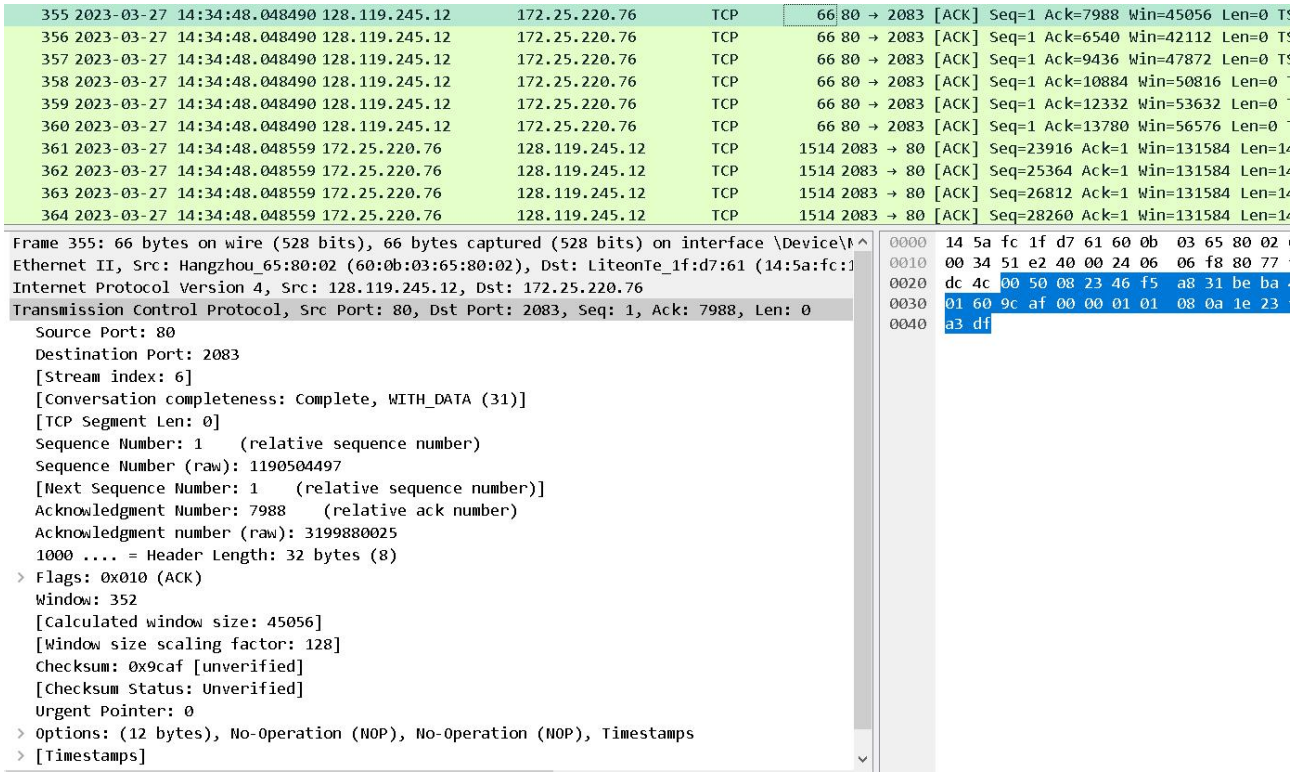
```
Frame 304: 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits) on interface \
Ethernet II, Src: LiteonTe_1f:d7:61 (14:5a:fc:1f:d7:61), Dst: JuniperN_f6:12:a0 (28:a2:4b:f
Internet Protocol Version 4, Src: 172.25.220.76, Dst: 128.119.245.12
Transmission Control Protocol, Src Port: 2083, Dst Port: 80, Seq: 5092, Ack: 1, Len: 1448
   Source Port: 2083
   Destination Port: 80
   [Stream index: 6]
   [Conversation completeness: Complete, WITH_DATA (31)]
   [TCP Segment Len: 1448]
   Sequence Number: 5092    (relative sequence number)
   Sequence Number (raw): 3199877129
   [Next Sequence Number: 6540    (relative sequence number)]
   Acknowledgment Number: 1    (relative ack number)
   Acknowledgment number (raw): 1190504497
   1000 .... = Header Length: 32 bytes (8)
 > Flags: 0x010 (ACK)
   Window: 514
   [Calculated window size: 131584]
   [Window size scaling factor: 256]
   Checksum: 0x719a [unverified]
   [Checksum Status: Unverified]
   Urgent Pointer: 0
 > Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
 > [Timestamps]
```

```
0020  f5 0c 08 23 00 50 be ba  40 09 46 f5 a8
0030  02 02 71 9a 00 00 01 01  08 0a 1d 26 a3
0040  e1 dc 74 68 69 6e 6b 20  79 6f 75 20 63
0050  64 20 6d 61 6e 61 67 65  20 69 74 3f 29
0060  41 6e 64 20 77 68 61 74  0d 0a 61 6e 20
0070  6f 72 61 6e 74 20 6c 69  74 74 6c 65 20
0080  6c 20 73 68 65 27 6c 6c  20 74 68 69 6e
0090  65 20 66 6f 72 20 61 73  6b 69 6e 67 21
00a0  6f 2c 20 69 74 27 6c 6c  0d 0a 6e 65 76
00b0  64 6f 20 74 6f 20 61 73  6b 3a 20 20 70
00c0  61 70 73 20 49 20 73 68  61 6c 6c 20 73
00d0  69 74 20 77 72 69 74 74  65 6e 20 75 70
00e0  6d 65 77 68 65 72 65 2e  27 0d 0a 0d 0a
00f0  6f 77 6e 2c 20 64 6f 77  6e 2c 20 64 6f
0100  20 20 54 68 65 72 65 20  77 61 73 20 6e
0110  69 6e 67 20 65 6c 73 65  20 74 6f 20 64
0120  73 6f 20 41 6c 69 63 65  20 73 6f 6f 6e
0130  65 67 61 6e 20 74 61 6c  6b 69 6e 67 20
0140  69 6e 2e 20 20 60 44 69  6e 61 68 27 6c
0150  69 73 73 20 6d 65 20 76  65 72 79 20 6d
0160  20 74 6f 2d 6e 69 67 68  74 2c 20 49 0d
0170  6f 75 6c 64 20 74 68 69  6e 6b 21 27 20
0180  69 6e 61 68 20 77 61 73  20 74 68 65 20
0190  2e 29 20 20 60 49 20 68  6f 70 65 20 74
01a0  27 6c 6c 20 72 65 6d 65  6d 62 65 72 0d
01b0  72 20 73 61 75 63 65 72  20 6f 66 20 6d
01c0  20 61 74 20 74 65 61 2d  74 69 6d 65 2e
```


356 2023-03-27 14:34:48.048490 128.119.245.12 172.25.220.76 TCP 66 80 → 2083 [ACK] Seq=1 Ack=6540 Win=42112 Len=0
357 2023-03-27 14:34:48.048490 128.119.245.12 172.25.220.76 TCP 66 80 → 2083 [ACK] Seq=1 Ack=9436 Win=47872 Len=0
358 2023-03-27 14:34:48.048490 128.119.245.12 172.25.220.76 TCP 66 80 → 2083 [ACK] Seq=1 Ack=10884 Win=50816 Len=0
359 2023-03-27 14:34:48.048490 128.119.245.12 172.25.220.76 TCP 66 80 → 2083 [ACK] Seq=1 Ack=12332 Win=53632 Len=0
360 2023-03-27 14:34:48.048490 128.119.245.12 172.25.220.76 TCP 66 80 → 2083 [ACK] Seq=1 Ack=13780 Win=56576 Len=0
361 2023-03-27 14:34:48.048559 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=23916 Ack=1 Win=131584 Len=
362 2023-03-27 14:34:48.048559 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=25364 Ack=1 Win=131584 Len=
363 2023-03-27 14:34:48.048559 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=26812 Ack=1 Win=131584 Len=
364 2023-03-27 14:34:48.048559 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=28260 Ack=1 Win=131584 Len=
Frame 356: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF
Ethernet II, Src: Hangzhou_65:80:02 (60:0b:03:65:80:02), Dst: LiteonTe_1f:d7:61 (14:5a:fc:1f:
Internet Protocol Version 4, Src: 128.119.245.12, Dst: 172.25.220.76
Transmission Control Protocol, Src Port: 80, Dst Port: 2083, Seq: 1, Ack: 6540, Len: 0
Source Port: 80
Destination Port: 2083
[Stream index: 6]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 0]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 1190504497
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 6540 (relative ack number)
Acknowledgment number (raw): 3199878577
1000 .... = Header Length: 32 bytes (8)
Flags: 0x010 (ACK)
Window: 329
[Calculated window size: 42112]
[Window size scaling factor: 128]
Checksum: 0xa26e [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
0000 14 5a fc 1f d7 61 60 0b 03 65 80 02
0010 00 34 51 e1 40 00 24 06 06 f9 80 77
0020 dc 4c 00 50 08 23 46 f5 a8 31 be ba
0030 01 49 a2 6e 00 00 01 01 08 0a 1e 23
0040 a3 df

## 区段六：


305 2023-03-27 14:34:47.770563 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=6540 Ack=1 Win=131584 Len=1448
306 2023-03-27 14:34:47.770563 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=7988 Ack=1 Win=131584 Len=1448
307 2023-03-27 14:34:47.770563 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=9436 Ack=1 Win=131584 Len=1448
308 2023-03-27 14:34:47.770563 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=10884 Ack=1 Win=131584 Len=1448
309 2023-03-27 14:34:47.770563 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=12332 Ack=1 Win=131584 Len=1448
rame 305: 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits) on interface \
thernet II, Src: LiteonTe_1f:d7:61 (14:5a:fc:1f:d7:61), Dst: JuniperN_f6:12:a0 (28:a2:4b:f
nternet Protocol Version 4, Src: 172.25.220.76, Dst: 128.119.245.12
ransmission Control Protocol, Src Port: 2083, Dst Port: 80, Seq: 6540, Ack: 1, Len: 1448
Source Port: 2083
Destination Port: 80
[Stream index: 6]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 1448]
Sequence Number: 6540 (relative sequence number)
Sequence Number (raw): 3199878577
[Next Sequence Number: 7988 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 1190504497
1000 .... = Header Length: 32 bytes (8)
Flags: 0x010 (ACK)
Window: 514
[Calculated window size: 131584]
[Window size scaling factor: 256]
Checksum: 0xe950 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
[Timestamps]
0000 28 a2 4b f6 12 a0 14 5a fc 1f d7 61 08
0010 05 dc 5e c0 40 00 80 06 98 71 ac 19 dc
0020 f5 0c 08 23 00 50 be ba 45 b1 46 f5 a8
0030 02 02 e9 50 00 00 01 01 08 0a 1d 26 a3
0040 e1 dc 20 41 6c 69 63 65 20 6c 69 6b 65
0050 65 20 77 69 6e 64 2c 20 61 6e 64 20 77
0060 6a 75 73 74 20 69 6e 20 74 69 6d 65 20
0070 68 65 61 72 20 69 74 0d 0a 73 61 79 2c
0080 20 69 74 20 74 75 72 6e 65 64 20 61 20
0090 6e 65 72 2c 20 60 4f 68 20 6d 79 20 65
00a0 20 61 6e 64 20 77 68 69 73 6b 65 72 73
00b0 6f 77 20 6c 61 74 65 0d 0a 69 74 27 73
00c0 74 74 69 6e 67 21 27 20 20 53 68 65 20
00d0 20 63 6c 6f 73 65 20 62 65 68 69 6e 64
00e0 20 77 68 65 6e 20 73 68 65 20 74 75 72
00f0 20 74 68 65 0d 0a 63 6f 72 6e 65 72 2c
0100 74 20 74 68 65 20 52 61 62 62 69 74 20
0110 20 6e 6f 20 6c 6f 6e 67 65 72 20 74 6f
0120 20 73 65 65 6e 3a 20 20 73 68 65 20 66
0130 64 0d 0a 68 65 72 73 65 6c 66 20 69 6e
0140 6c 6f 6e 67 2c 20 6c 6f 77 20 68 61 6c
0150 77 68 69 63 68 20 77 61 73 20 6c 69 74
0160 20 62 79 20 61 20 72 6f 77 20 6f 66 20
0170 70 73 0d 0a 68 61 6e 67 69 6e 67 20 66
0180 20 74 68 65 20 72 6f 6f 66 2e 0d 0a 0d
0190 54 68 65 72 65 20 77 65 72 65 20 64 6f

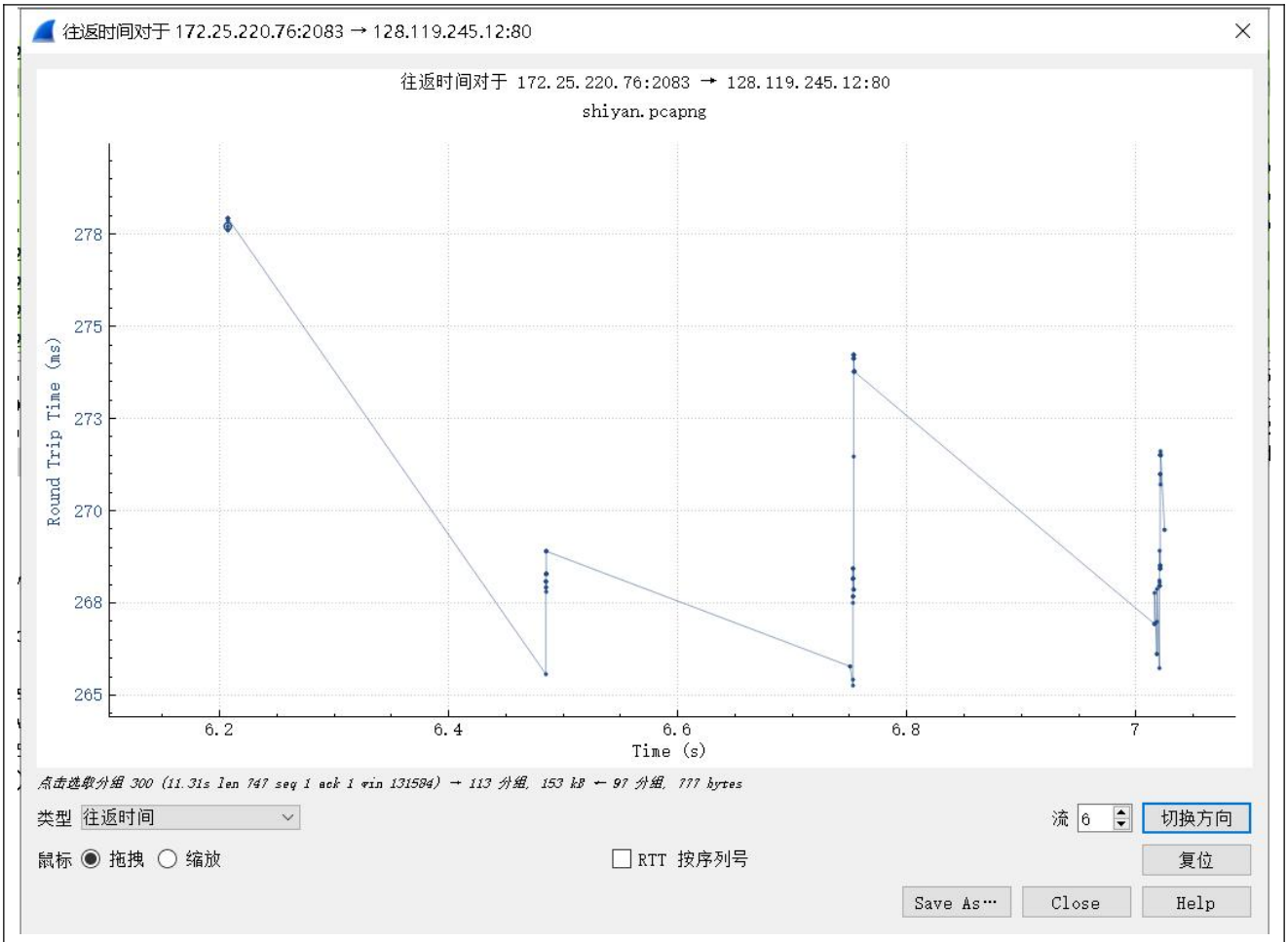



RTT 分别为 0.277708，0.277606，0.277606，0.277856，0.277927，0.277927。

由公式：

$$EstimatedRTT = (1-a) \times EstimatedRTT + a \times SampleRTT, a = 0.125$$

有 EstimatedRTT 为 0.277708，0.277695，0.277650，0.277676，0.277707，0.2777735。




8. 前六个段长度分别为 747，1448，1448，1448，1448，1448。

9.

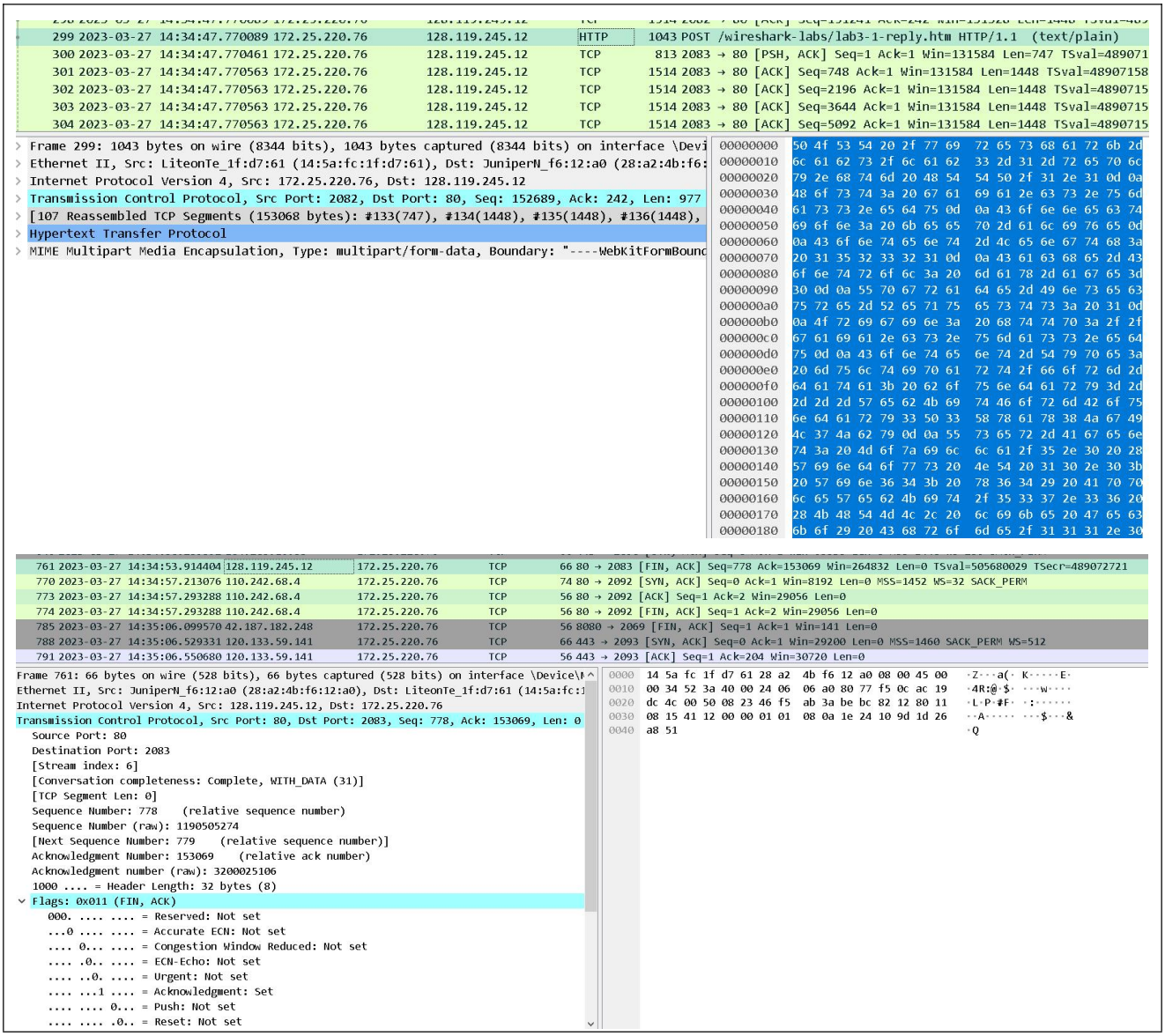
| | | | | | | |
|---|---|---|---|---|---|---|
| 344 | 2023-03-27 14:34:48.048169 | 128.119.245.12 | 172.25.220.76 | TCP | 66 | 80 → 2083 [ACK] Seq=1 Ack=748 Win=30464 Len=0 |
| 345 | 2023-03-27 14:34:48.048169 | 128.119.245.12 | 172.25.220.76 | TCP | 66 | 80 → 2083 [ACK] Seq=1 Ack=3644 Win=36352 Len=0 |
| 346 | 2023-03-27 14:34:48.048169 | 128.119.245.12 | 172.25.220.76 | TCP | 66 | 80 → 2083 [ACK] Seq=1 Ack=2196 Win=33408 Len=0 |
| 347 | 2023-03-27 14:34:48.048280 | 172.25.220.76 | 128.119.245.12 | TCP | 1514 | 2083 → 80 [ACK] Seq=13780 Ack=1 Win=131584 Len=1 |
| 348 | 2023-03-27 14:34:48.048280 | 172.25.220.76 | 128.119.245.12 | TCP | 1514 | 2083 → 80 [ACK] Seq=15228 Ack=1 Win=131584 Len=1 |
| 349 | 2023-03-27 14:34:48.048280 | 172.25.220.76 | 128.119.245.12 | TCP | 1514 | 2083 → 80 [PSH, ACK] Seq=16676 Ack=1 Win=131584 |
| 350 | 2023-03-27 14:34:48.048280 | 172.25.220.76 | 128.119.245.12 | TCP | 1514 | 2083 → 80 [ACK] Seq=18124 Ack=1 Win=131584 Len=1 |
| 351 | 2023-03-27 14:34:48.048280 | 172.25.220.76 | 128.119.245.12 | TCP | 1514 | 2083 → 80 [ACK] Seq=19572 Ack=1 Win=131584 Len=1 |

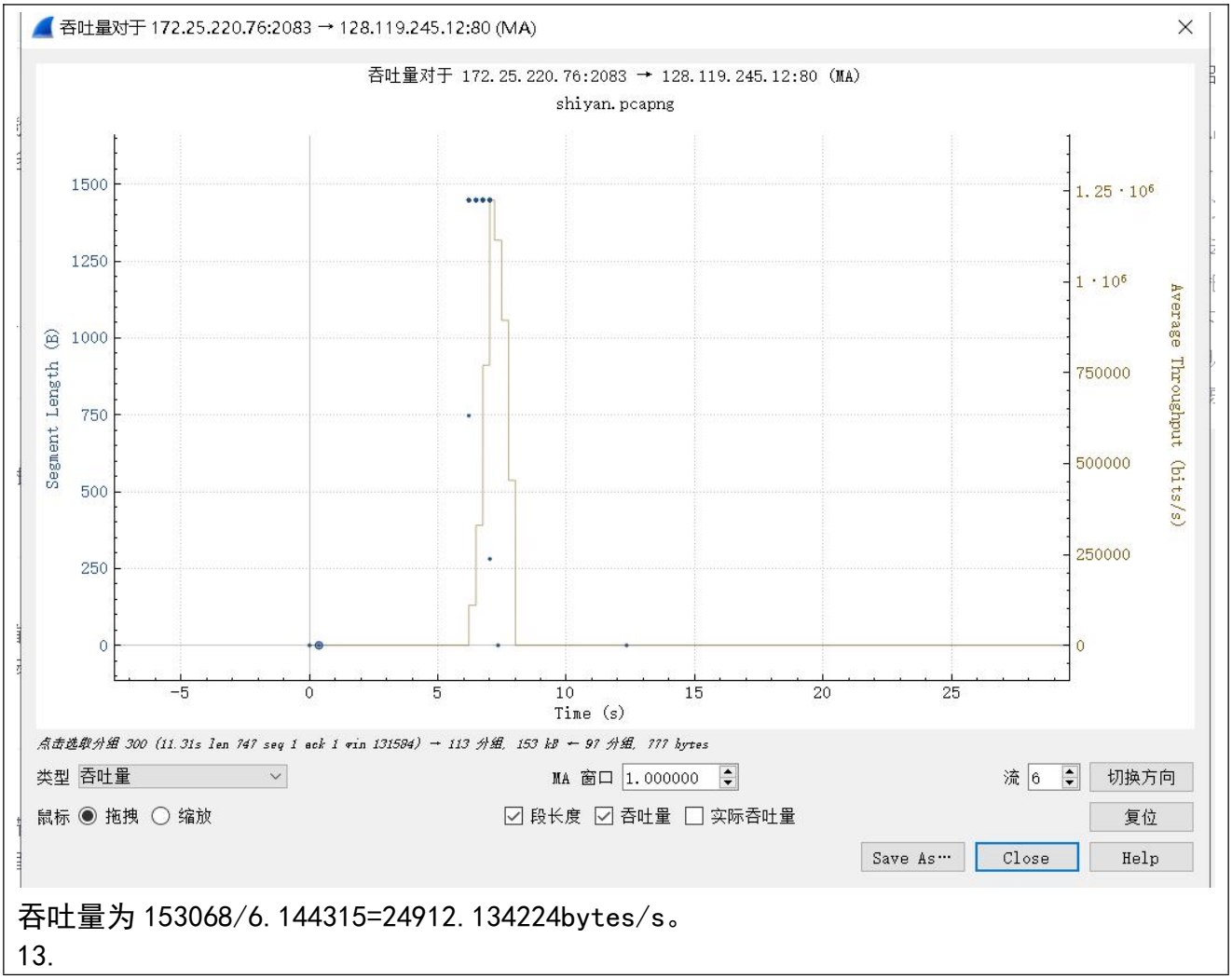
最小为 30464，后逐渐增大，如果缺少缓冲空间，会限制发送方发送数据。

10. 并没有发现重传的段，因为序列号是一直在增大的，而没有出现重复。

11. 确认数据量大部分为 1448bytes，少部分小于 1448bytes，TCP 是累计确认的，所以再发送序号之前的都是确认收到了的。

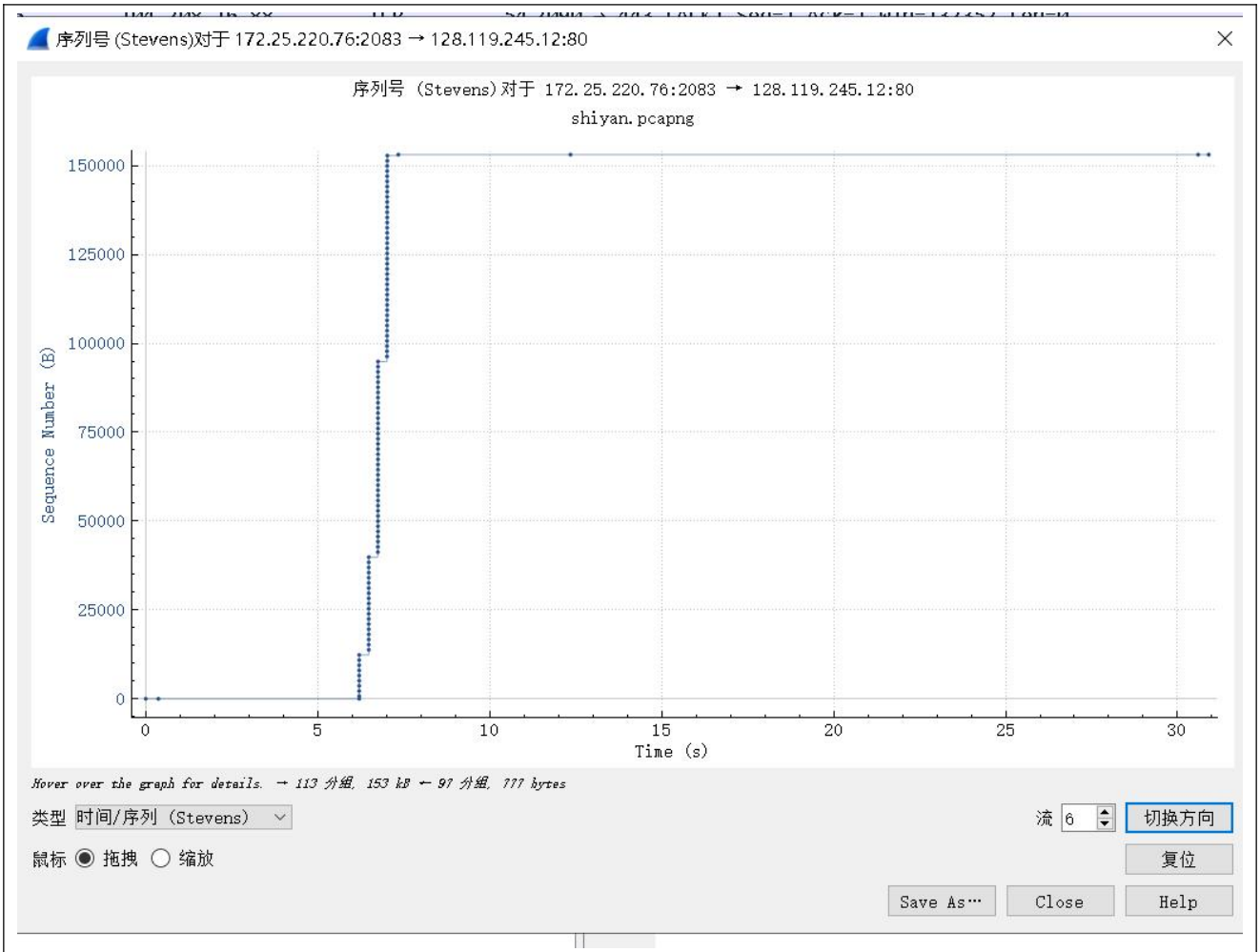
12.


299 2023-03-27 14:34:47.770089 172.25.220.76 128.119.245.12 HTTP 1043 POST /wireshark-labs/lab3-1-reply.htm HTTP/1.1 (text/plain)
300 2023-03-27 14:34:47.770461 172.25.220.76 128.119.245.12 TCP 813 2083 → 80 [PSH, ACK] Seq=1 Ack=1 Win=131584 Len=747 TSval=489071
301 2023-03-27 14:34:47.770563 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=748 Ack=1 Win=131584 Len=1448 TSval=48907158
302 2023-03-27 14:34:47.770563 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=2196 Ack=1 Win=131584 Len=1448 TSval=4890715
303 2023-03-27 14:34:47.770563 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=3644 Ack=1 Win=131584 Len=1448 TSval=4890715
304 2023-03-27 14:34:47.770563 172.25.220.76 128.119.245.12 TCP 1514 2083 → 80 [ACK] Seq=5092 Ack=1 Win=131584 Len=1448 TSval=4890715
Frame 299: 1043 bytes on wire (8344 bits), 1043 bytes captured (8344 bits) on interface \Devi
Ethernet II, Src: LiteonTe_1f:d7:61 (14:5a:fc:1f:d7:61), Dst: JuniperN_f6:12:a0 (28:a2:4b:f6:
Internet Protocol Version 4, Src: 172.25.220.76, Dst: 128.119.245.12
Transmission Control Protocol, Src Port: 2082, Dst Port: 80, Seq: 152689, Ack: 242, Len: 977
[107 Reassembled TCP Segments (153068 bytes): #133(747), #134(1448), #135(1448), #136(1448),
Hypertext Transfer Protocol
MIME Multipart Media Encapsulation, Type: multipart/form-data, Boundary: "----WebKitFormBound
00000000 50 4f 53 54 20 2f 77 69 72 65 73 68 61 72 6b 2d
00000010 6c 61 62 73 2f 6c 61 62 33 2d 31 2d 72 65 70 6c
00000020 79 2e 68 74 6d 20 48 54 54 50 2f 31 2e 31 0d 0a
00000030 48 6f 73 74 3a 20 67 61 69 61 2e 63 73 2e 75 6d
00000040 61 73 73 2e 65 64 75 0d 0a 43 6f 6e 6e 65 63 74
00000050 69 6f 6e 3a 20 6b 65 65 70 2d 61 6c 69 76 65 0d
00000060 0a 43 6f 6e 74 65 6e 74 2d 4c 65 6e 67 74 68 3a
00000070 20 31 35 32 33 32 31 0d 0a 43 61 63 68 65 2d 43
00000080 6f 6e 74 72 6f 6c 3a 20 6d 61 78 2d 61 67 65 3d
00000090 30 0d 0a 55 70 67 72 61 64 65 2d 49 6e 73 65 63
000000a0 75 72 65 2d 52 65 71 75 65 73 74 73 3a 20 31 0d
000000b0 0a 4f 72 69 67 69 6e 3a 20 68 74 74 70 3a 2f 2f
000000c0 67 61 69 61 2e 63 73 2e 75 6d 61 73 73 2e 65 64
000000d0 75 0d 0a 43 6f 6e 74 65 6e 74 2d 54 79 70 65 3a
000000e0 20 6d 75 6c 74 69 70 61 72 74 2f 66 6f 72 6d 2d
000000f0 64 61 74 61 3b 20 62 6f 75 6e 64 61 72 79 3d 2d
00000100 2d 2d 2d 57 65 62 4b 69 74 46 6f 72 6d 42 6f 75
00000110 6e 64 61 72 79 33 50 33 58 78 61 78 38 4a 67 49
00000120 4c 37 4a 62 79 0d 0a 55 73 65 72 2d 41 67 65 6e
00000130 74 3a 20 4d 6f 7a 69 6c 6c 61 2f 35 2e 30 20 28
00000140 57 69 6e 64 6f 77 73 20 4e 54 20 31 30 2e 30 3b
00000150 20 57 69 6e 36 34 3b 20 78 36 34 29 20 41 70 70
00000160 6c 65 57 65 62 4b 69 74 2f 35 33 37 2e 33 36 20
00000170 28 4b 48 54 4d 4c 2c 20 6c 69 6b 65 20 47 65 63
00000180 6b 6f 29 20 43 68 72 6f 6d 65 2f 31 31 31 2e 30
761 2023-03-27 14:34:53.914404 128.119.245.12 172.25.220.76 TCP 66 80 → 2083 [FIN, ACK] Seq=778 Ack=153069 Win=264832 Len=0 TSval=505680029 TSecr=489072721
770 2023-03-27 14:34:57.213076 110.242.68.4 172.25.220.76 TCP 74 80 → 2092 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=1452 WS=32 SACK_PERM
773 2023-03-27 14:34:57.293288 110.242.68.4 172.25.220.76 TCP 56 80 → 2092 [ACK] Seq=1 Ack=2 Win=29056 Len=0
774 2023-03-27 14:34:57.293288 110.242.68.4 172.25.220.76 TCP 56 80 → 2092 [FIN, ACK] Seq=1 Ack=2 Win=29056 Len=0
785 2023-03-27 14:35:06.099570 42.187.182.248 172.25.220.76 TCP 56 8080 → 2069 [FIN, ACK] Seq=1 Ack=1 Win=141 Len=0
788 2023-03-27 14:35:06.529331 120.133.59.141 172.25.220.76 TCP 66 443 → 2093 [SYN, ACK] Seq=0 Ack=1 Win=29200 Len=0 MSS=1460 SACK_PERM WS=512
791 2023-03-27 14:35:06.550680 120.133.59.141 172.25.220.76 TCP 56 443 → 2093 [ACK] Seq=1 Ack=204 Win=30720 Len=0
Frame 761: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\N
Ethernet II, Src: JuniperN_f6:12:a0 (28:a2:4b:f6:12:a0), Dst: LiteonTe_1f:d7:61 (14:5a:fc:1
Internet Protocol Version 4, Src: 128.119.245.12, Dst: 172.25.220.76
Transmission Control Protocol, Src Port: 80, Dst Port: 2083, Seq: 778, Ack: 153069, Len: 0
Source Port: 80
Destination Port: 2083
[Stream index: 6]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 0]
Sequence Number: 778 (relative sequence number)
Sequence Number (raw): 1190505274
[Next Sequence Number: 779 (relative sequence number)]
Acknowledgment Number: 153069 (relative ack number)
Acknowledgment number (raw): 3200025106
1000 .... = Header Length: 32 bytes (8)
Flags: 0x011 (FIN, ACK)
000. .... .... = Reserved: Not set
...0 .... .... = Accurate ECN: Not set
.... 0... .... = Congestion Window Reduced: Not set
.... .0.. .... = ECN-Echo: Not set
.... ..0. .... = Urgent: Not set
.... ...1 .... = Acknowledgment: Set
.... .... 0... = Push: Not set
.... .... .0.. = Reset: Not set
0000 14 5a fc 1f d7 61 28 a2 4b f6 12 a0 08 00 45 00
0010 00 34 52 3a 40 00 24 06 06 a0 80 77 f5 0c ac 19
0020 dc 4c 00 50 08 23 46 f5 ab 3a be bc 82 12 80 11
0030 08 15 41 12 00 00 01 01 08 0a 1e 24 10 9d 1d 26
0040 a8 51




吞吐量为 153068/6.144315=24912.134224bytes/s。

13.




慢启动是从连接开始时，发送窗口大小呈指数型增长，上图中没用发现慢启动结束和拥塞避免。

结论分析与体会：

学习了 TCP 的相关内容，通过查看具体的 TCP 封包中的内容，对 TCP 以及 TCP 的三次握手过程有了进一步的了解，同时也学会如何使用 Wireshark 来查看具体的一些数据，如吞吐量和时间序列等。